THE DETERMINANT OF A PERFECT COMPLEX OF TOR AMPLITUDE IN TWO DEGREES IS A FUNCTOR TO INVERTIBLE MODULES, WITH A CANONICAL ELEMENT CARRYING ISOMORPHISMS

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Allowed standard prerequisites (editorial).

- The determinant of a two-term complex of finite projective modules as det(L^0) tensor det(L^{-1})^{tensor -1}. used at proof lines 37160-37162. Is the object the functor is built from.
- The three conditions on the maps of a good diagram established earlier in the section. used at proof lines 37173-37181 and 37198-37200. Are what make det(c^bullet) o det(b^bullet)^{-1} a sensible definition.
- The homotopic-surjections lemma comparing the determinants of two maps that agree up to homotopy. invoked at proof line 37210. Gives independence of det(a^bullet) from the chosen good diagram in the identity case.
- Representation of morphisms from a bounded complex of finite projective modules by homotopy classes of maps of complexes. invoked at proof lines 37156-37159. Lets the construction be carried out on good complexes rather than abstractly.

Source reference. Original lemma 064B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7860–7860. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 36985: sources/stacks-more-algebra/pinned-original-source.tex, source lines 36985–36985. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

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Source reference. Original lemma 37108: sources/stacks-more-algebra/pinned-original-source.tex, source lines 37108–37108. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 37016: sources/stacks-more-algebra/pinned-original-source.tex, source lines 37016–37016. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 18889: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18889–18889. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. The constructions above determine a functor*

$$\det : \left\{ \begin{array}{l} \text{category of perfect complexes} \\ \text{with tor amplitude in } [-1, 0] \\ \text{morphisms are isomorphisms} \end{array} \right\} \longrightarrow \left\{ \begin{array}{l} \text{category of invertible modules} \\ \text{morphisms are isomorphisms} \end{array} \right\}$$

Moreover, given a rank 0 perfect object L of $D(R)$ with tor-amplitude in $[-1, 0]$ there is a canonical element $\delta(L) \in \det(L)$ such that for any isomorphism $a : L \to K$ in $D(R)$ we have $\det(a)(\delta(L)) = \delta(K)$.

Proof. By Lemma 18889 every object of the source category may be represented by a complex

$$L^\bullet = \ldots \to 0 \to L^{-1} \to L^0 \to 0 \to \ldots$$

with L^{-1} and L^0 finite projective R-modules. Let us temporarily call a complex of this type good. By Derived Categories, Lemma 064B morphisms between good complexes in the derived category are homotopy classes of maps of complexes. Thus we may work with good complexes and we can use the determinant $\det(L^\bullet) = \det(L^0) \otimes \det(L^{-1})^{\otimes -1}$ we investigated above.

Let $a^\bullet : L^\bullet \to K^\bullet$ be a morphism of good complexes which is an isomorphism in $D(R)$, i.e., a quasi-isomorphism. We say that

$$\begin{array}{ccc} L^\bullet & \xrightarrow{\quad a^\bullet \quad} & K^\bullet \\ & {\scriptstyle b^\bullet} \nwarrow \quad \nearrow {\scriptstyle c^\bullet} & \\ & M^\bullet & \end{array}$$

is a good diagram if it commutes up to homotopy and $b^\bullet$ and $c^\bullet$ satisfy conditions (1), (2), (3) above. Whenever we have such a diagram it makes sense to define

$$\det(a^\bullet) = \det(c^\bullet) \circ \det(b^\bullet)^{-1}$$

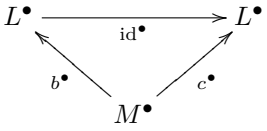
where $\det(c^\bullet)$ and $\det(b^\bullet)$ are the isomorphisms constructed in the text above. We will show that good diagrams always exist and that the resulting map $\det(a^\bullet)$ is independent of the choice of good diagram.

Existence of good diagrams for a quasi-isomorphism $a^\bullet : L^\bullet \to K^\bullet$ of good complexes. Choose a surjection $p : R^{\oplus n} \to K^{-1}$. Then we can consider the new good complex

$$M^\bullet = \ldots \to 0 \to L^{-1} \oplus R^{\oplus n} \xrightarrow{d \oplus 1} L^0 \oplus R^{\oplus n} \to 0 \to \ldots$$

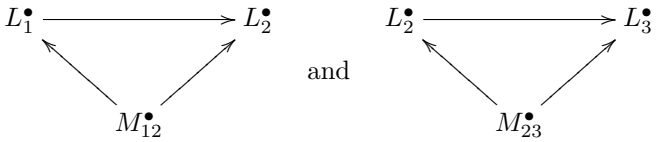
with the projection map $b^\bullet : M^\bullet \to L^\bullet$ and the map $c^\bullet : M^\bullet \to K^\bullet$ using $a^{-1} \oplus p$ in degree -1 and using $a^0 \oplus d \circ p$ in degree 0. The maps $b^\bullet : M^\bullet \to L^\bullet$ and $c^\bullet : M^\bullet \to K^\bullet$ satisfy conditions (1), (2), (3) above and we get a good diagram.

Suppose that we have a good diagram

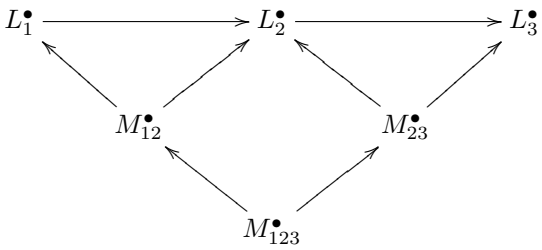


Then by Lemma 37016 we see that $\det(c^\bullet) = \det(b^\bullet)$. Thus we see that $\det(\mathrm{id}^\bullet) = \mathrm{id}$ is independent of the choice of good diagram.

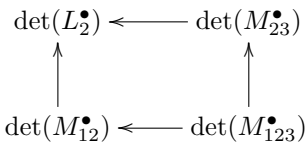
Before we prove independence in general, we think about composition. Suppose we have quasi-isomorphisms $L_1^\bullet \to L_2^\bullet$ and $L_2^\bullet \to L_3^\bullet$ of good complexes and good diagrams



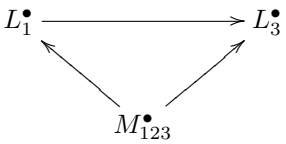
We can extend this to a diagram



where $M_{123}^\bullet \to M_{12}^\bullet$ and $M_{123}^\bullet \to M_{23}^\bullet$ have properties (1), (2), (3) and the square in the diagram commutes: we can just take $M_{123}^n = M_{12}^n \times_{L_2^n} M_{23}^n$. Then Lemma 37108 shows that



commutes. A diagram chase shows that the composition $\det(L_1^\bullet) \to \det(L_2^\bullet) \to \det(L_3^\bullet)$ of the maps associated to the two good diagrams using $M_{12}^\bullet$ and $M_{23}^\bullet$ is equal to the map associated to the good diagram



Thus if we can show that these maps are independent of choices, then the composition law is satisfied too and we obtain our functor.

Independence. Let a quasi-isomorphism $a^\bullet : L^\bullet \to K^\bullet$ of good complexes be given. Choose an inverse quasi-isomorphism $b^\bullet : K^\bullet \to L^\bullet$. Setting $L_1^\bullet = L$, $L_2^\bullet = K^\bullet$ and $L_3^\bullet = L^\bullet$ may fix our choice of good diagram for $b^\bullet$ and consider varying good diagrams for $a^\bullet$. Then the result of the previous paragraphs is that no matter what choices, the composition always equals the identity map on $\det(L^\bullet)$. This clearly proves independence of those choices.

The statement on canonical elements follows immediately from Lemma 36985 and our construction. □